rate and depth to match metabolic demands. Additional input to the control center is provided by sensors in the aorta and carotid arteries that monitor blood levels of $O_2$ as well as $CO_2$ (via blood pH).

? *How does air in the lungs differ from the fresh air that enters the body during inspiration?*

CONCEPT **42.7**

### Adaptations for gas exchange include pigments that bind and transport gases *(pp. 947–949)*

- In the lungs, gradients of partial pressure favor the net diffusion of $O_2$ into the blood and $CO_2$ out of the blood. The opposite situation exists in the rest of the body. **Respiratory pigments** such as hemocyanin and hemoglobin bind $O_2$, greatly increasing the amount of $O_2$ transported by the circulatory system.
- Evolutionary adaptations enable some animals to satisfy extraordinary $O_2$ demands. Deep-diving mammals stockpile $O_2$ in blood and other tissues and deplete it slowly.

? *How are the roles of a respiratory pigment and an enzyme similar?*

## TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

### Levels 1-2: Remembering/Understanding

**1.** Which of the following respiratory systems is independent from a fluid-based circulatory system?
(A) the lungs of a vertebrate
(B) the gills of a fish
(C) the tracheal system of an insect
(D) the skin of an earthworm

**2.** Blood returning to the mammalian heart in a pulmonary vein drains first into the
(A) left atrium.
(B) right atrium.
(C) left ventricle.
(D) right ventricle.

**3.** Pulse is a direct measure of
(A) blood pressure.
(B) stroke volume.
(C) cardiac output.
(D) heart rate.

**4.** When you hold your breath, which of the following blood gas changes first leads to the urge to breathe?
(A) rising $O_2$
(B) falling $O_2$
(C) rising $CO_2$
(D) falling $CO_2$

**5.** One feature that amphibians and humans have in common is
(A) the number of heart chambers.
(B) a complete separation of circuits for circulation.
(C) the number of circuits for circulation.
(D) a low blood pressure in the systemic circuit.

### Levels 3-4: Applying/Analyzing

**6.** A molecule of $CO_2$ released into the blood in your left toe can be exhaled from your nose without passing through which of the following structures?
(A) the pulmonary vein.
(B) the trachea.
(C) the right atrium.
(D) the right ventricle.

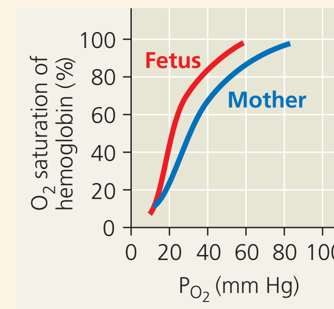
**7.** Compared with the interstitial fluid that bathes active muscle cells, blood reaching these cells in arterioles has a
(A) higher $P_{O_2}$.
(B) higher $P_{CO_2}$.
(C) greater bicarbonate concentration.
(D) lower pH.

### Levels 5-6: Evaluating/Creating

**8. DRAW IT** Plot blood pressure against time for one cardiac cycle in humans, drawing separate lines for the pressure in the aorta, the left ventricle, and the right ventricle. Below the time axis, add a vertical arrow pointing to the time when you expect a peak in atrial blood pressure.

**9. EVOLUTION CONNECTION** One opponent of the movie monster Godzilla is Mothra, a mothlike creature with a wingspan of several dozen meters. The largest known insects were Paleozoic dragonflies with half-meter wingspans. Focusing on respiration and gas exchange, explain why giant insects are improbable.

**10. SCIENTIFIC INQUIRY • INTERPRET THE DATA** The hemoglobin of a human fetus differs from adult hemoglobin. Compare the dissociation curves of the two hemoglobins in the graph at right. Describe how they differ, and propose a hypothesis to explain the benefit of this difference.




**11. SCIENCE, TECHNOLOGY, AND SOCIETY** Hundreds of studies have linked smoking with cardiovascular and lung disease. According to most health authorities, smoking is the leading cause of preventable, premature death in the United States. What are some arguments in favor of a total ban on cigarette advertising? What are arguments in opposition? Do you favor or oppose such a ban? Explain.

**12. WRITE ABOUT A THEME: INTERACTIONS** Some athletes prepare for competition at sea level by sleeping in a tent in which $P_{O_2}$ is kept low. When climbing high peaks, some mountaineers breathe from bottles of pure $O_2$. In a short essay (100–150 words), relate these behaviors to the mechanism of $O_2$ transport in the human body and to physiological interactions with our gaseous environment.

**13. SYNTHESIZE YOUR KNOWLEDGE**

The diving bell spider (*Argyroneta aquatica*) stores air underwater in a net of silk. Explain why this adaptation could be more advantageous than having gills, taking into account differences in gas exchange media and gas exchange organs among animals.

*For selected answers, see Appendix A.*